THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 13: **Our Environment** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these constitutes a food chain? 1

(a) grass, goat and human **(b)** wolf, lion and tiger
(c) grass, wheat and mango **(d)** goat, cow and elephant

2. Which group contains only biodegradable items? 1

(a) grass, flowers and leather **(b)** cake, wood and glass
(c) grass, wood and plastic **(d)** fruit peels, cake and polythene

3. The organisms at the first trophic level are: 1

(a) herbivores **(b)** producers **(c)** carnivores **(d)** decomposers

4. A student states: “Removing all decomposers from an ecosystem has no effect.” The correct response is: 1

(a) Correct
(b) Correct for ponds
(c) Incorrect — wastes would pile up and nutrients would not be recycled
(d) Cannot be determined

5. For the source of energy in all ecosystems, Student A says the Sun; Student B says decomposers. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. Green plants in a terrestrial ecosystem capture about ___ of the sunlight falling on leaves. 1

(a) 1% **(b)** 10% **(c)** 50% **(d)** 90%

7. Ozone is a molecule made of: 1

(a) one oxygen atom **(b)** oxygen and carbon
(c) three oxygen atoms **(d)** two oxygen atoms

8. Which of these is a decomposer? 1

(a) grasshopper **(b)** frog **(c)** grass **(d)** bacteria

9. If a herbivore gets 1000 J, the energy available to the carnivore eating it is about: 1

(a) 900 J **(b)** 100 J **(c)** 1000 J **(d)** 10 J

10. An aquarium is an example of a/an: 1

(a) natural ecosystem **(b)** artificial ecosystem
(c) food chain **(d)** biome

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What are trophic levels? 2

12. Why is it necessary to clean an aquarium regularly though it is an ecosystem? 2

13. Name two abiotic components of an ecosystem. 2

14. A student says that replacing plastic cups with kulhads (clay cups) on a large scale is fully eco-friendly. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What will happen if we kill all the organisms in one trophic level? 3

16. Explain why the impact of removing all organisms in a trophic level differs for different levels. 3

17. What is the function of ozone in the upper atmosphere? How is it formed? 3

OR

How do our habits affect the environment with respect to waste? Give two changes we can make. 3

18. What are the advantages of reusable over disposable items (three)? 3

OR

Why do producers always occupy the first trophic level? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What is a food chain? 5
(ii) Give an example of an aquatic and a terrestrial food chain.
(iii) What would happen if decomposers were absent?

OR

(i) What is biological magnification? 5
(ii) How do pesticides enter the food chain?
(iii) Why does it affect humans most?
(iv) Name one chemical that shows it.

20. **Case study:** A city produces tonnes of garbage daily including plastic, food waste and glass. 5
(i) Classify the waste.
(ii) Suggest methods of disposal of each.
(iii) Why should plastic not be burnt?
(iv) What is meant by the 3 R's?

OR

(i) Explain the 10% law of energy flow with a diagram. 5
(ii) If 10 000 J is available at producers, how much energy reaches the tertiary consumer?
(iii) Why are there fewer animals at higher trophic levels?

END OF QUESTION PAPER — Total: 40 Marks